ASSIGNMENT-11.2

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BATCH-30

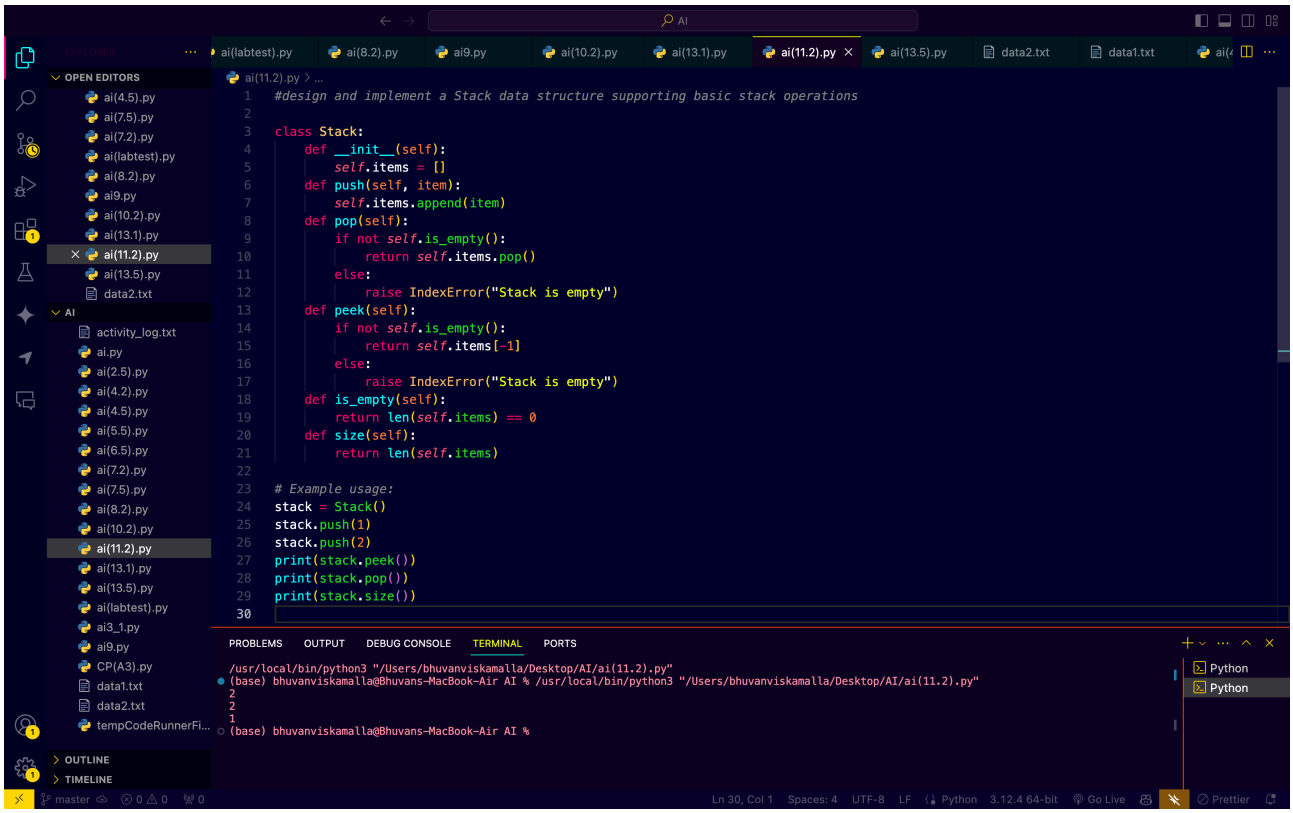
TASK-1:

PROMPT:

#Generate a python code to design and implement a Stack datastructure supporting basic
stack operations.

A Python Stack class supporting push, pop, peek, and empty-check operations with proper documentation.

CODE:



EXPECTED OUTPUT:
A Python Stack class supporting push, pop, peek, and empty-check operations with proper documentation.

OBSERVATION :

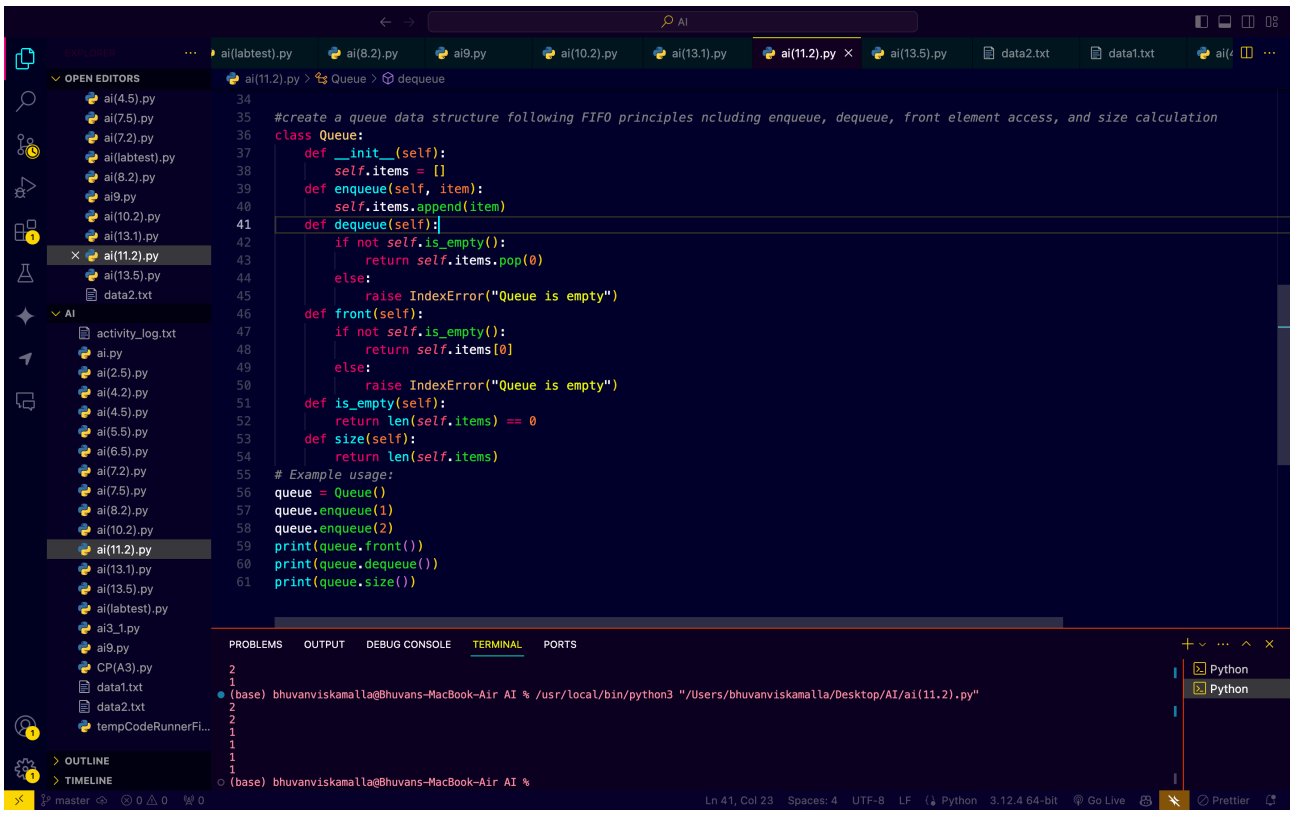
-It is observed that in the out put will get by implementing FIFO principles
-The program successfully demonstrated stack and handled empty stack conditions

TASK-2:

PROMPT:

#Generate a python code to create a Queue data structure following FIFO principles
A complete Queue implementation including enqueue, dequeue,
front element access, and size calculation

CODE:



Expected Output:

• A complete Queue implementation including enqueue, dequeue,
front element access, and size calculation

OBSERVATION:

-it was able to implement the queue and dequeue ,font, size

- The program successfully demonstrated the queue operations

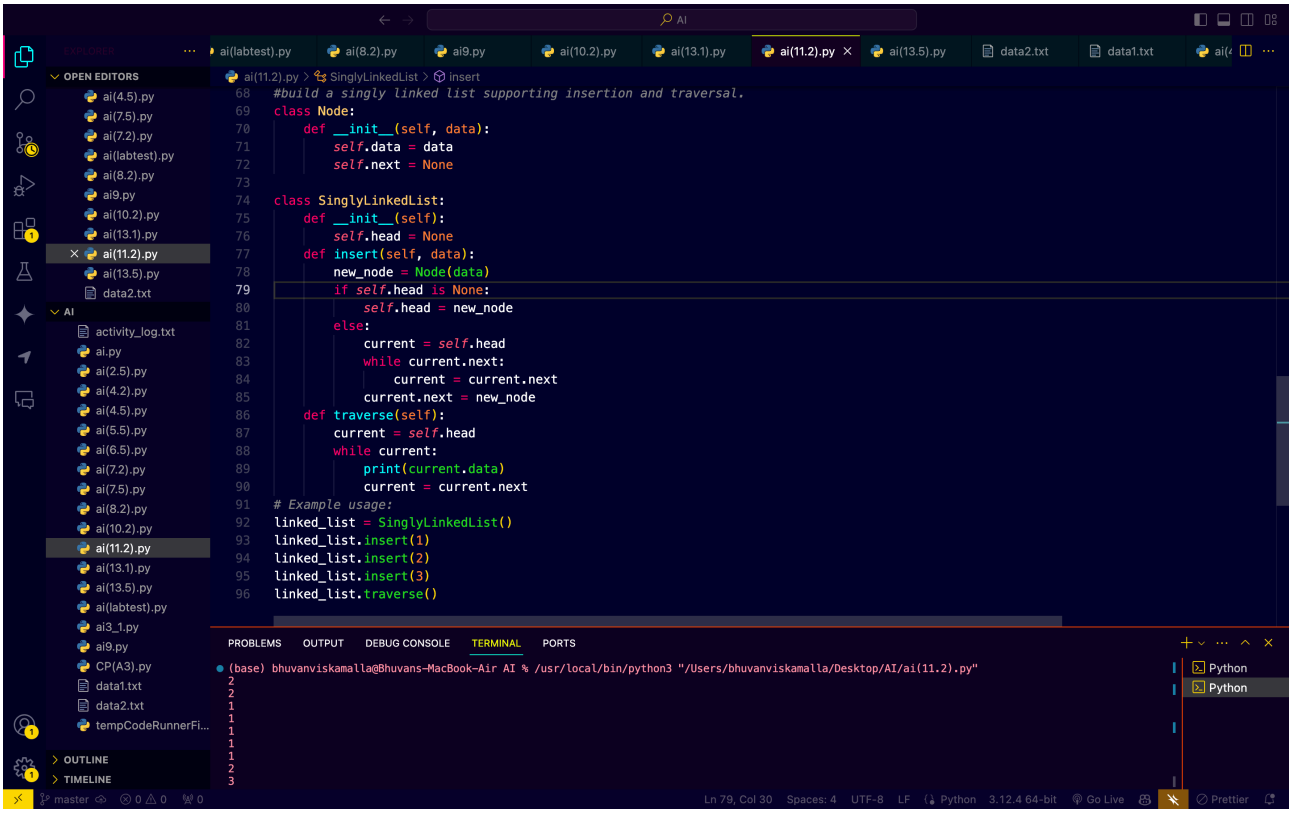
- TASK-3:

PROMPT:

Generate a python code build a singly linked list supporting insertion
and traversal.

Correctly functioning linked list with node creation, insertion
logic, and display functionality.

CODE:



Expected Output:

• Correctly functioning linked list with node creation, insertion
logic, and display functionality.

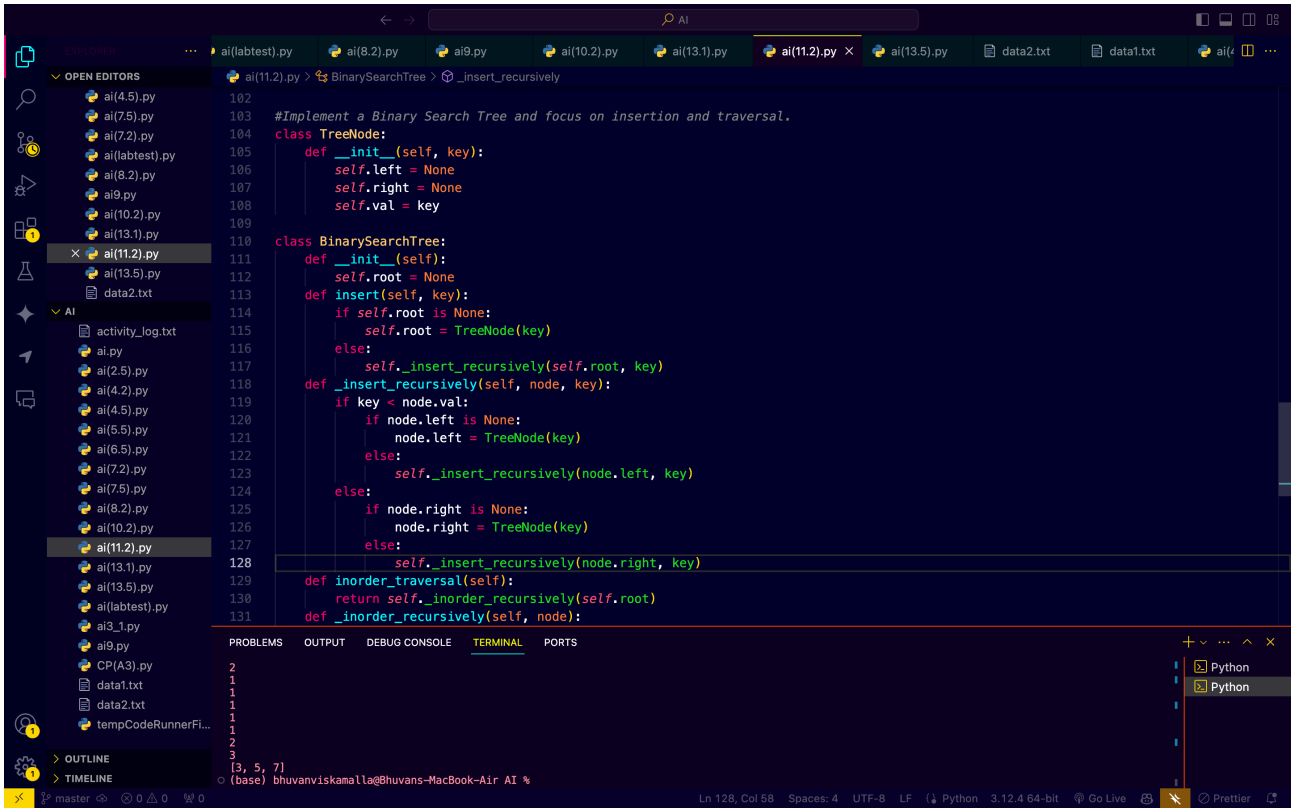
OBSERVATION:
-it was able to perform all the linked list operations
-it was getting the accurate expected output by the generated example usage code.

TASK-4:

PROMPT:

Generate a python code to get a Binary Search Tree with AI support focusing on
insertion and traversal.
#BST program with correct node insertion and in-order traversal
output.

CODE:



Expected Output:

- BST program with correct node insertion and in-order traversal

.

Output

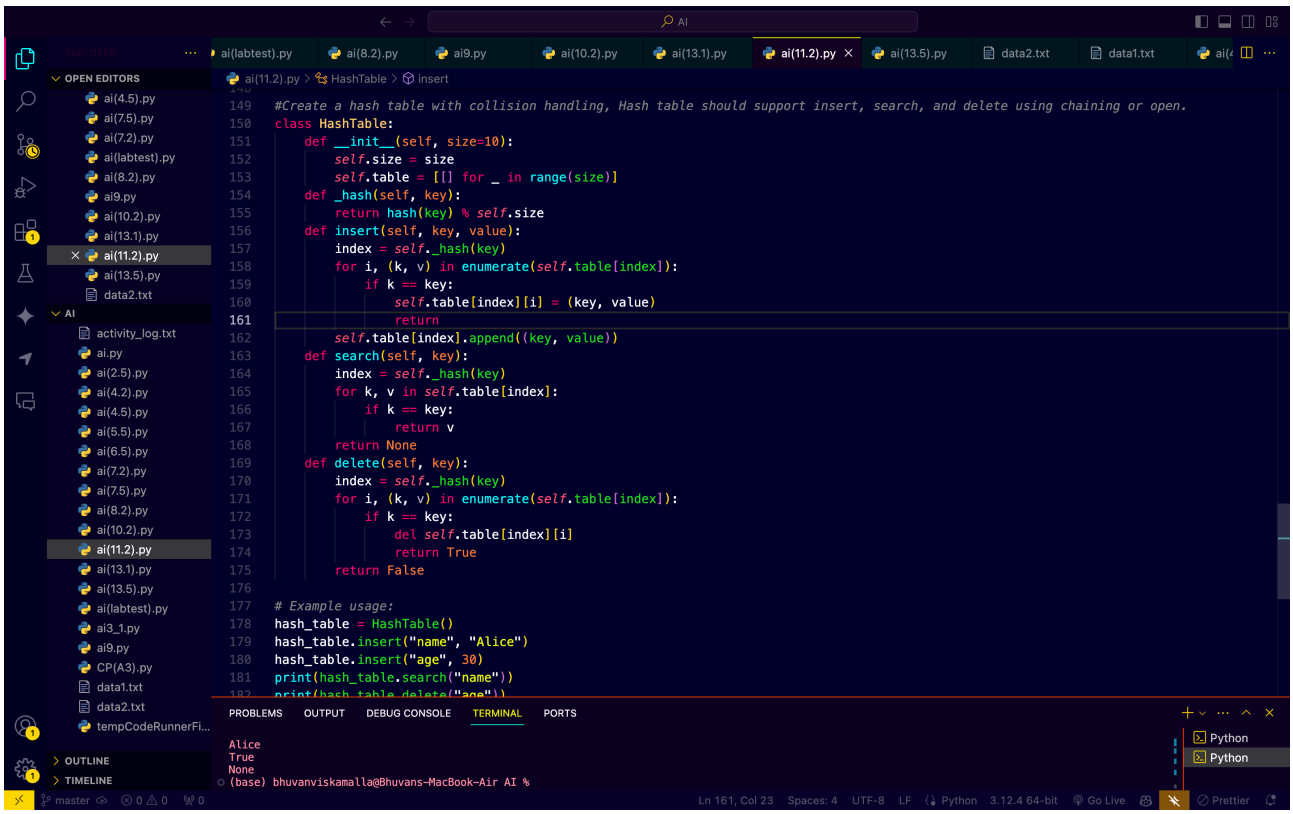
OBSERVATION:
-it was able to perform all the BinarySearch Tree operations
-it was getting the accurate expected output by the generated example usage code.

TASK-5

PROMPT:
Generate a python code to Create a hash table using AI with collision handling

#Hash table supporting insert, search, and delete using chaining
or open

CODE:



Expected Output:

• Hash table supporting insert, search, and delete using chaining
or open

OBSERVATION:

-it was able to perform all the Hash Table operations
-it was getting the accurate expected output by the generated example usage code.